MTH321 · Numerical Analysis of Partial Differential Equations

# Finite Difference Solution of a Poisson Problem with Mixed Boundary Data

Does the right-edge closure decide the accuracy?

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# Presentation Roadmap

How does the **right-edge boundary row** affect accuracy, matrix structure and computational cost?

**01**

## Problem

Mixed boundary data and two closures

**02**

## Theory

Why $O(h)$ versus $O(h^2)$?

**03**

## Verification

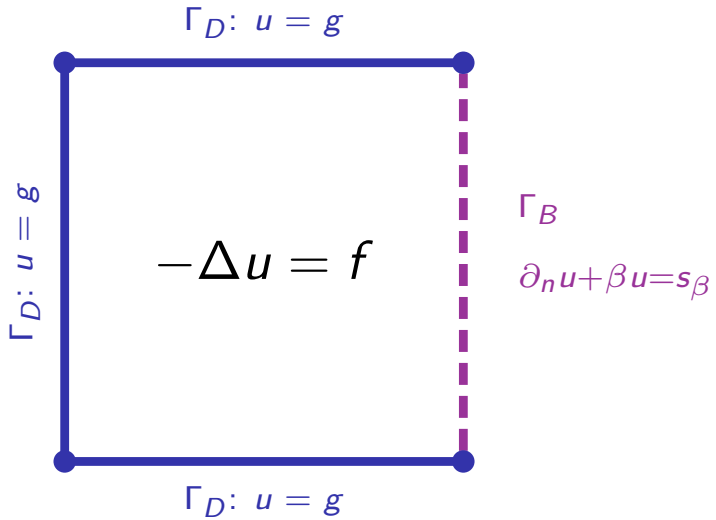
Can we trust the numerical results?

**04**

## Results

Convergence, error, cost and lessons

# Why Do Boundary Rows Matter?




- **Neumann** ($\beta = 0$): prescribed flux; insulated edge if $q = 0$
- **Robin** ($\beta = \mathrm{Bi} = HL/k$): convection to a fluid; $\beta = 2$ means $\mathrm{Bi} = 2$
- **Benchmark** $u^* = e^{x+y}$: exact data, so the error is known

Same five-point interior stencil — only the right-edge row is a modelling choice.

**Takeaway.** Real boundaries mix values, fluxes and convection; the closure is part of the method, so we test it.

# Two Right-Edge Closures

## FO: derivative row

Backward difference **replaces** the PDE row:

$$\left(\frac{1}{h}+\beta\right)U_{N,j}-\frac{1}{h}U_{N-1,j}=s_{\beta,j}$$

No tangential coupling

Boundary defect: $-\frac{h}{2}u_{xx}$

## G: ghost point, PDE row kept

Eliminate $U_{N+1,j}$ (ghost):

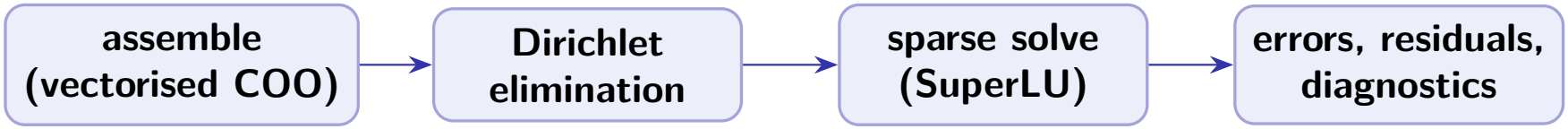
$$\left(\frac{4}{h^2}+\frac{2\beta}{h}\right)U_{N,j}-\frac{2}{h^2}U_{N-1,j}-\frac{U_{N,j-1}+U_{N,j+1}}{h^2}=f+\frac{2s_\beta}{h}$$

Boundary defect: $+\frac{h}{3}u_{xxx}$

Interior defect $O(h^2)$ · Boundary defects $O(h)$ · $N(N-1)$ unknowns

**Takeaway.** FO imposes a derivative row, while G keeps the PDE row. Their global performance may differ because of stability.

# From Equations to a Solver




- Nonzeros: FO $(N-1)(5N-6)$, G $+2(N-2)$
- Raw matrices are **nonsymmetric**: $\|A - A^T\|_F / \|A\|_F = 0.055$ at $N = 32$
- Weights $W$ (1 inside, $1/h$ for FO, $1/2$ for G) give $K = WA$ **symmetric positive definite**

- Implemented by us: stencils, elimination, norms, weights
- Delegated: SciPy sparse storage, SuperLU, GMRES, CG
- `run_all.py` regenerates every table and figure

**Takeaway.** Nonsymmetry is not hidden by the self-adjoint PDE: CG applies only to the weighted system, which we prove SPD.

# Why Does G Gain a Factor $h$?

| | boundary defect | stability estimate | max-norm error |
|---|---|---|---|
| FO | $\frac{M_2}{2}h$ | $\|V\|_\infty \le \frac{7}{2}(F_0 + S_0)$ | $O(h)$ |
| G | $\frac{M_3}{3}h + \frac{M_4}{12}h^2$ | $\|V\|_\infty \le \frac{7}{4}(2F_0 + hS_0)$ | $O(h^2)$ |

- Proof tools: **discrete comparison principle** and the barrier $\psi = 1 + x - x^2/4$; the matrices are inverse positive
- The estimate is uniform in $N$ and $\beta$: stability, not consistency, separates the two closures
- Interior defect is $O(h^2)$ for both closures

**Takeaway.** G's $O(h)$ boundary defect enters the error multiplied by $h$; FO's does not, so FO is $O(h)$ and G is $O(h^2)$.

# From Theory to Testable Predictions

1. FO error $\Theta(h)$, G error $O(h^2)$, in the maximum and the discrete $L^2$ norm
2. The largest error sits on the **right edge**
3. FO has a **lower bound** $C_\beta h$ ($C_0 = 0.305$, $C_2 = 0.186$): first order is real, not just an upper bound

| FO at $N = 128$ | lower bound | measured | upper bound |
|---|---|---|---|
| Neumann | $2.4 \times 10^{-3}$ | $6.9 \times 10^{-3}$ | $1.0 \times 10^{-1}$ |
| Robin | $1.5 \times 10^{-3}$ | $4.2 \times 10^{-3}$ | $1.0 \times 10^{-1}$ |

The measured error lies between the bounds; the upper bound is conservative (15–24 times too large).

**Takeaway.** Theory gives predictions we can test; the lower bound shows FO's first order is genuine, not a weakness of the proof.

# Four Independent Checks of the Equations

## 1. Hand-derived row

For $N = 4$, $j = 1$: G row entries $64 + 8\beta$, $-32$, $-16$.
Load: $(6 + 8\beta)e^{5/4} + 16e$.

## 2. Polynomial defects

Test function: $u = 1 + x^3 + y^4$.
$\tau_I = -2h^2$, $\tau_{\mathrm{FO}} = -3h + h^2$,
$\tau_{\mathrm{G}} = 2h - 2h^2$.

## 3. Dense-system check

Uneliminated system: all Dirichlet and ghost unknowns kept.
All 16 configurations agree to $10^{-11}$.

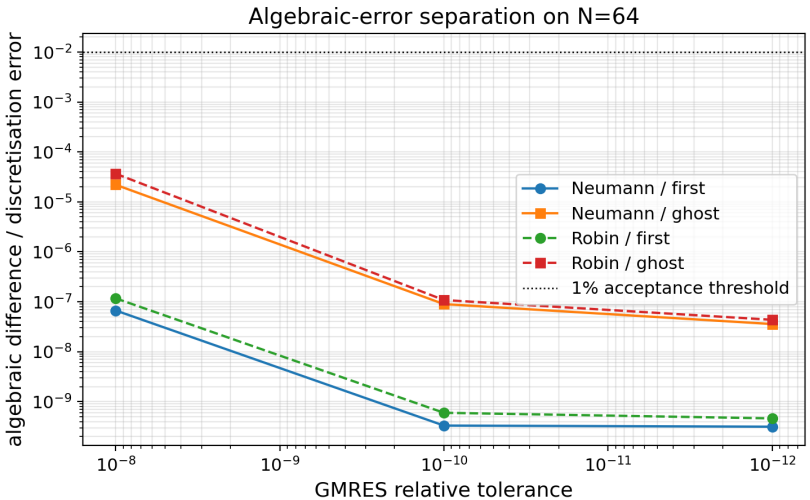
## 4. Independent P1 FEM

Independent reference: scikit-fem.
FEM error order: 2.00.
FDM–FEM difference: 1.00 (FO), 2.00 (G).

16 automated tests · Asymmetric test function catches $x \leftrightarrow y$ swaps.

**Takeaway.** Four independent checks agree: the assembled matrix matches the intended equations.

# Is the Linear Solve Trustworthy?

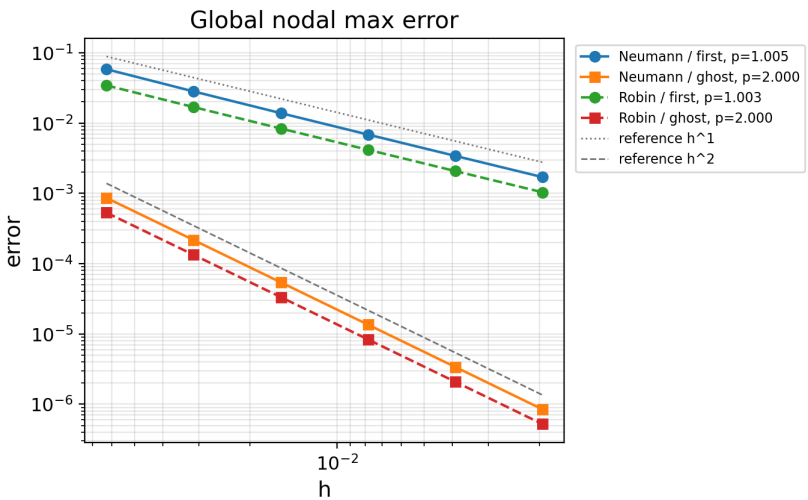


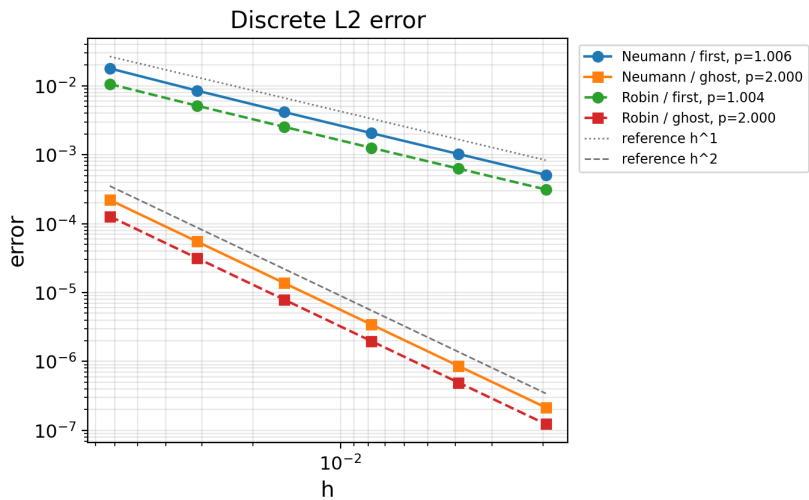

GMRES/direct difference relative to discretisation error ($N = 64$); all below the 1% line

- Direct solve: componentwise residual $\leq 2.3 \times 10^{-12}$ over 24 runs
- $\|A^{-1}\|_\infty$ stays in $[0.09, 0.51]$ for all $h$; $\kappa_\infty \sim h^{-2}$
- Estimated forward error $\leq 2.8 \times 10^{-6}$ of the discretisation error
- ILU-GMRES and weighted CG agree: ratio $\leq 3.7 \times 10^{-5}$
- Jacobi-GMRES at rtol $10^{-8}$ fails for G ($7.4 \times 10^{-2}$); passes at $10^{-10}$

**Takeaway.** Solver error is far below discretisation error, but the stopping tolerance must be chosen per closure: G needs a stricter one.

# First Versus Second Order

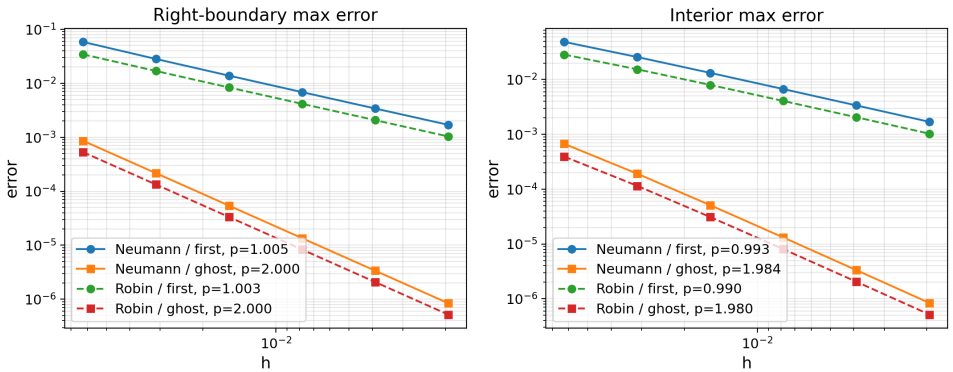



| | FO $p_\infty$ | G $p_\infty$ | $E_\infty$(FO) at $h = \frac{1}{512}$ | $E_\infty$(G) |
|---|---|---|---|---|
| Neumann | 1.005 | 2.000 | $1.7 \times 10^{-3}$ | $8.5 \times 10^{-7}$ |
| Robin | 1.003 | 2.000 | $1.0 \times 10^{-3}$ | $5.2 \times 10^{-7}$ |

**Takeaway.** FO converges at order 1.00 and G at 2.00; at $h = 1/512$ G is about 2000 times more accurate with only 0.08% more nonzeros.

# Where Does the Error Live?

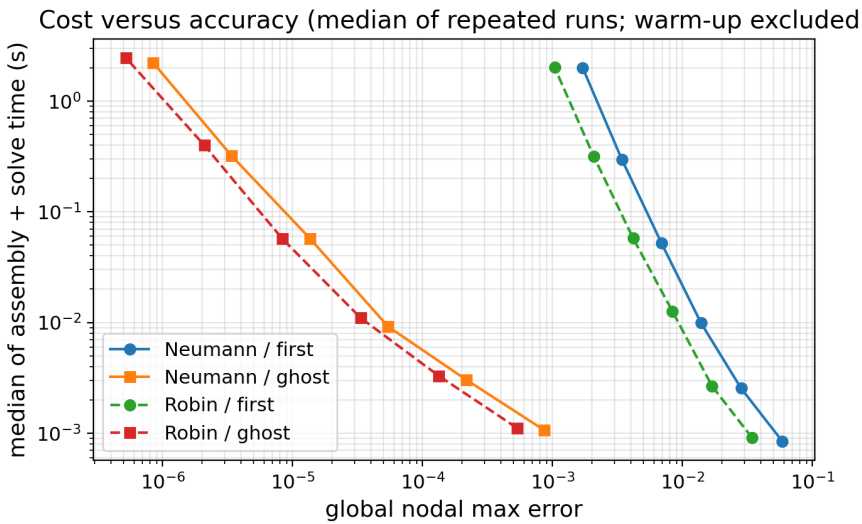



Maximum error on the right edge and in the interior

- Global maximum = **right-edge** maximum in every run
- FO interior order $\approx 1$: the boundary defect pollutes the whole field
- G interior fit 1.98; pairwise orders rise from 1.8 to 1.99 as the maximiser moves to $x = 1$

**Takeaway.** A second-order interior stencil does not give a second-order solution: the boundary row limits global accuracy.

# What Does the Accuracy Cost?




Assembly plus solve time versus nodal error ($N = 4, \ldots, 512$)

- Targets $5 \times 10^{-3} \ldots 10^{-5}$: FO reaches only the two loosest within $N \leq 512$
- Target $2 \times 10^{-3}$ (Neumann): FO needs $N = 512$ ($\approx 2.0$ s), G needs $N = 16$ ($\approx 1$ ms)
- At equal $N = 512$ the cost is similar: 2.0 s (FO) vs 2.2 s (G)

**Takeaway.** G is cheaper only because it reaches each target on a much coarser grid: accuracy per unknown, not speed per unknown.

# What We Learned

## Findings

- The right-edge **row**, not the interior stencil, decides the order
- Weights give the stability constants; FO is provably first order
- Four independent checks plus solver cross-checks support the numbers

## Limits

- Smooth manufactured benchmark; corner singularities not covered
- No matching lower bound for G
- Timings belong to one machine; this is verification, not experimental validation

**Takeaway.** Verify the boundary row, not just the interior stencil: it controls accuracy, structure and cost.

# Backup: Stability Proof Idea

- **Comparison principle**: if $V \geq 0$ on the Dirichlet nodes, $L_h V \geq 0$ inside and the boundary row is $\geq 0$, then $V \geq 0$ everywhere
- **Barrier** $\psi = 1 + x - x^2/4 \in [1, \frac{7}{4}]$: $\quad L_h\psi = \frac{1}{2}, \;\; B_h^\beta \psi \geq \frac{1}{2}, \;\; G_h^\beta \psi \geq \frac{1}{h}$
- Compare $a\psi \pm V$ with $a = 2(F_0 + S_0)$ (FO) or $a = 2F_0 + hS_0$ (G): $|V| \leq \frac{7}{4} a$
- Error equation $Ae = -\tau$ then gives $\|e^{\mathrm{FO}}\| = O(h)$, $\|e^{\mathrm{G}}\| = O(h^2)$

**Takeaway.** The factor $h$ in G comes from $G\psi \geq 1/h$: the barrier is large on the boundary row.

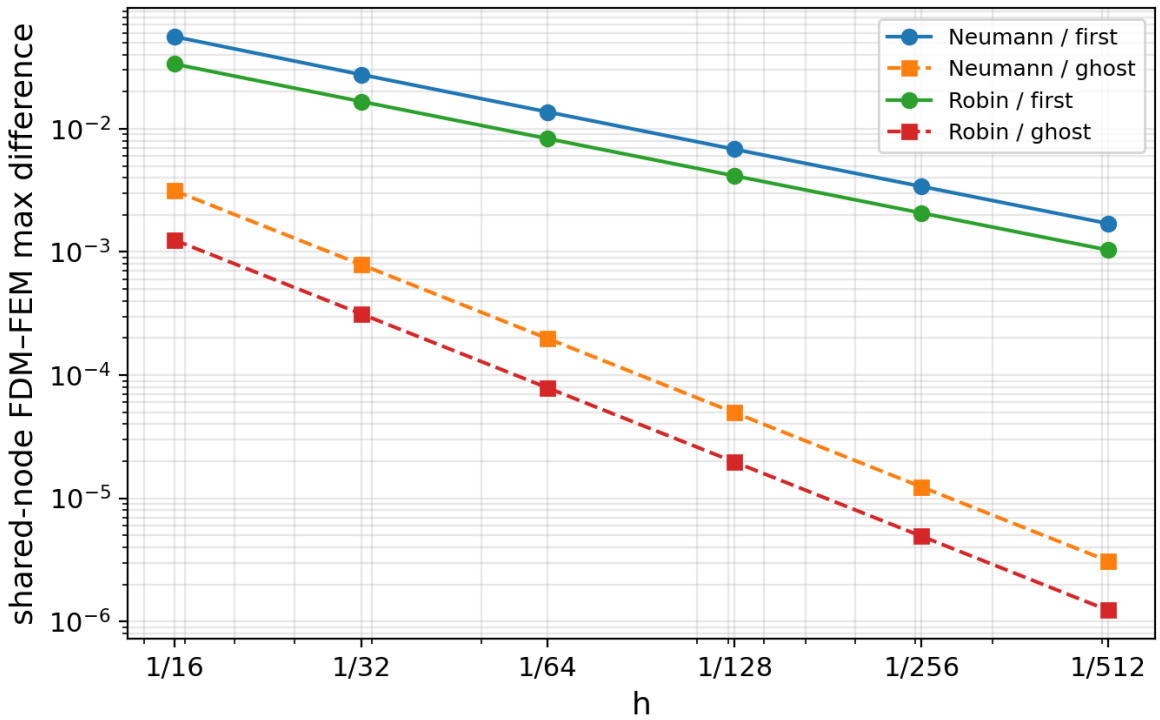
# Backup: Conditioning of the Direct Solve

| | $\|A^{-1}\|_\infty$ | $\kappa_\infty$ $(N = 64)$ | estimate$/E_\infty$ (max) |
|---|---|---|---|
| FO | 0.29–0.51 | $9.5 \times 10^3$–$1.6 \times 10^4$ | $2.3 \times 10^{-8}$ |
| G | 0.09–0.11 | $3.0 \times 10^3$–$3.7 \times 10^3$ | $2.8 \times 10^{-6}$ |

ILU-GMRES: ratio $\leq 3.7 \times 10^{-5}$ at rtol $10^{-8}$; weighted CG: $\leq 3.1 \times 10^{-5}$.

**Takeaway.** $\|A^{-1}\|_\infty$ is bounded independently of $h$, consistent with the proved stability constants.

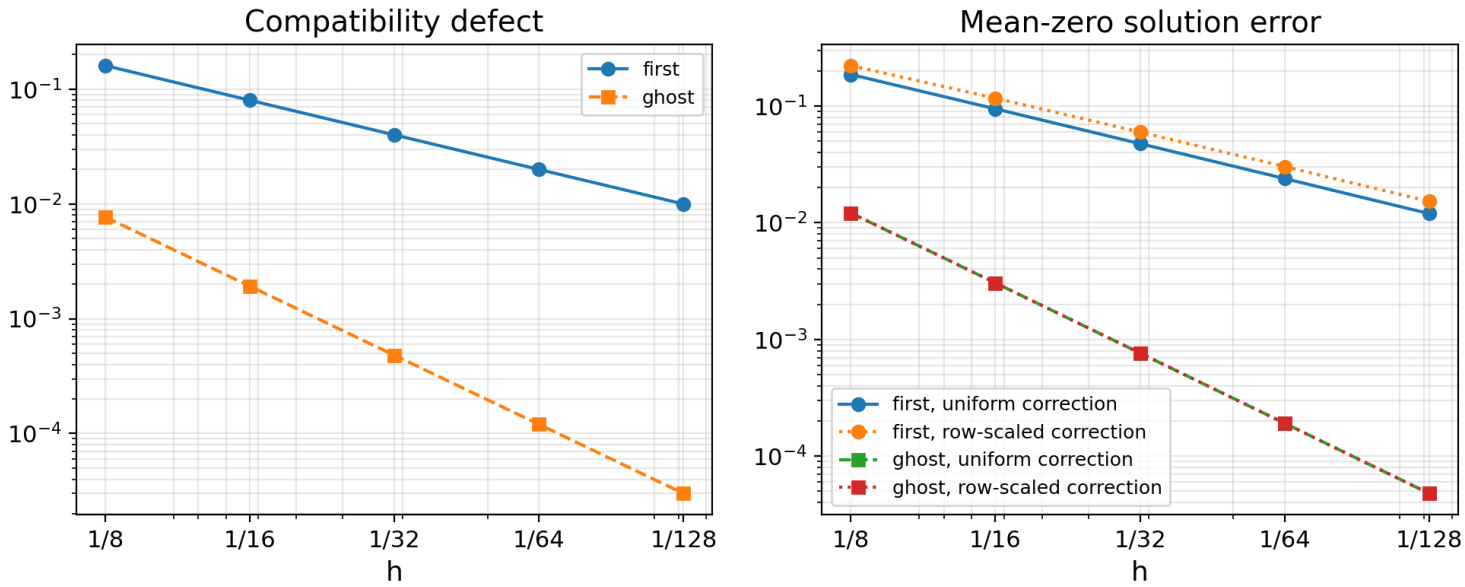
# Backup: Independent FEM Oracle




Maximum FDM–FEM difference at shared vertices, $N = 16, \ldots, 512$

**Takeaway.** FO–FEM decays at order 1.00 and G–FEM at 2.00: two discretisations of one weak problem agree.

# Backup: Pure-Neumann Extension




**Takeaway.** The FO compatibility defect is $O(h)$, G's is $O(h^2)$; a row-scaled correction does not improve FO.

# Backup: Reproducibility

- `code/run_all.py` regenerates all CSV data, LaTeX tables and figures
- 16 automated tests: rows, corners, defects, energy identity, FEM oracle, null space
- Run recorded in `data/run_metadata.json`: Python 3.13.5, NumPy 2.1.3, SciPy 1.15.3
- Direct solver: SuperLU through `spsolve`; timings: median after warm-up

**Takeaway.** Every number in the report comes from a generated file, not from manual entry.